

DAVID W. KASTNER

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Education

Ph.D. Bioengineering	MIT, Cambridge, MA	2025
	GPA 5.0 Graduate Student Spotlight	
B.S. Biophysics	BYU, Provo, UT	2019
	GPA 4.0 Honors <i>Summa Cum Laude</i> , Valedictorian Speech	

Fellowships

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|---|------|
| • Siebel Scholars Fellowship and Award (USD 35,000) | 2025 |
| • MIT Whitehead Fellowship | 2021 |
| • Vice Chancellor's Inclusive Excellence Fellowship | 2020 |
| • National Science Foundation Graduate Research Fellowship (NSF GRFP) | 2019 |
| • Tolero Pharmaceuticals Fellowship | 2018 |
| • Inspired Learning University Fellowship | 2018 |
| • Simmons Center for Cancer Research Fellowship (SCCR) | 2017 |

Prestigious Awards

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| • ACS Chemical Computing Group (CCG) Research Excellence Award | 2024 |
| • Alfred P. Sloan Exemplary Mentorship Award (UCEM) | 2019 |
| • Elva Pedersen Jorgenson Scholarship | 2019 |
| • Warren Rollins and Murdell Hull Scholarship | 2019 |
| • National Hispanic Fund Scholarship (HSF) | 2018 |
| • Lieutenant Governor's Community Service Award | 2018 |
| • National President's Volunteer Service Award | 2018 |
| • National Honor Society Phi Kappa Phi Outstanding Student Award | 2018 |
| • Karl G. Maeser Scholarship | 2018 |
| • University Undergraduate Research Award (URA) | 2018 |
| • National Barry Goldwater Scholarship | 2017 |
| • National American Chemical Society Scholars Award (ACS) | 2017 |
| • Full-Ride Merit-Based Scholarship | 2017 |

Research Experience

Massachusetts Institute of Technology (MIT) – Cambridge, MA Sept 2020 – Current

Co-Advisors: Dr. Heather J. Kulik and Dr. Forest White

- Bioengineering PhD candidate researching how non-covalent interactions drive enzyme catalysis
- 26 peer-reviewed manuscripts on proteins, enzymes, catalysis, and simulation
- Bridged ML and simulation by generating high-fidelity data sets for model training and validation
- Lead software developer: [QuantumPDB](#), [pyQMMM](#), [PyEF](#), [QuantumAllostery](#), [MolecuLearn](#)
- Delivered four research presentations at national conferences
- Mentored 11 junior researchers (6 undergraduate and 5 graduate students)
- Invited to act as a peer reviewer for 24 manuscripts

Pfizer Pharmaceutical Internship – Cambridge, MA Sept 2023 – Sept 2024

Internship | Machine Learning Computational Sciences (MLCS) Group

- Fine-tuned generative models specifically for predicting molecular glue binding poses
- Developed ML and physics-based workflows for predicting molecular-glue degrader binding
- Benchmarked generative models AF2, Boltz-1, DiffDock-L, RFAA, Chai-1, and Protenix for glues
- 200 million molecule structure-based high-throughput virtual screen of molecular glues

Huntsman Cancer Institute (HCI) – Salt Lake City, UT

Dec 2018 – Aug 2020

Oncological Sciences Department

- Full-time cancer computational and wet-lab biologist researching small cell lung cancer (SCLC)
- Collaborated with biologists, chemists, and bioinformaticians to publish high-impact publications

National Institutes of Health (NIH) – Bethesda, MD

Jul 2018 – Sept 2018

Advisor: Dr. Nico Tjandra

- Biophysics internship requiring electron microscopy, light scattering, NMR, circular dichroism

Dana-Farber/Harvard Cancer Center (DFCI/HCC) – Boston, MA

Apr 2018 – Jul 2018

Advisor: Dr. Haribabu Arthanari

- Biophysics internship requiring NMR, protein expression, biochemical pathway engineering

Simmons Center for Cancer Research (SCCR) – Provo, UT

May 2014 – Apr 2018

Advisor: Dr. Steven L. Castle

- Organic synthesis and computationally modelling (QM/MM) of anticancer peptides
- Engineered a potent anticancer peptide in collaboration with *Bristol-Myers Squibb* (BMS).

Technical Skills**Simulation and Modeling Tools**

- **Molecular Dynamics:** AMBER, OpenMM, CHARMM, PLUMED
- **Quantum Chemistry:** ORCA, TeraChem, Psi4, GFN2-xTB, CREST
- **Machine Learning:** PyTorch, JAX, DiffDock-L, RFDiffusion, AlphaFold, Boltz-1
- **Scripting & Pipelines:** Python, Bash, SLURM, SGE
- **Modeling:** Schrödinger Suite (FEP+, Maestro, Glide, LiveDesign), PyRosetta, AutoDock, MOE

Specializations

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| • Spanish (bilingual) | • Lead Python software developer (GitHub) |
| • Computational chemistry | • Enhanced sampling |
| • Machine learning and generative models | • Metadynamics, OPES, TPS |
| • High-performance computing (HPC) | • End-point free energy calculations |
| • Hybrid QM/MM multiscale simulations | • Computer-Aided Drug Discovery (CADD) |
| • Quantum mechanical (QM) calculations | • Molecular visualization expert |
| • Density functional theory (DFT) | • Molecular docking |
| • Molecular dynamics (MD) simulations | • High-throughput virtual screening (HTVS) |

Certifications[Fundamentals of Deep Learning](#) issued by NVIDIA

- PyTorch, CNNs, data augmentation, transfer learning, and natural language processing

[Fundamentals of Accelerated Computing with CUDA Python](#) issued by NVIDIA

- CUDA kernels, memory management, parallelization, GPU acceleration, and performance profiling

[Molecular Modeling in Drug Discovery](#) issued by Schrödinger

- LigPrep, Epik, Glide, and Ligand Designer in Maestro; ADMET profiling in LiveDesign

[Free Energy Calculations for Drug Design with FEP+](#) issued by Schrödinger

- REST sampling, λ windows, core-matching, and FEP-graph building with Maestro and Desmond

[Designing Quality Ligand Libraries](#) issued by Schrödinger

- Strucconvert, LigPrep, LibraryAnalysis, canvasSearch, drug-likeness and REOS/PAINS filtering

[Advanced Visualization with PyMol 3](#) issued by Schrödinger[Computational Chemistry School](#) at Novartis Cambridge, MA[Coleman Leadership Program](#) issued by Cornell Engineering and SLOAN

Publications

Highlighted Front Cover Publications ([Google Scholar](#)) ([ORCID](#)) ([GitHub](#))

1. [\[Front Cover\]](#) **Kastner, D.W.**; Nandy, A.; Mehmood, R.; Kulik, H.J. Mechanistic Insights into Substrate Positioning That Distinguishes Non-heme Fe(II)/ α -Ketoglutarate-Dependent Halogenases and Hydroxylases. *ACS Catal.* **2023**, 13 (4), 2489-2501. [DOI](#).
2. [\[Front Cover\]](#) Duan, C.; Nandy, A.; Terrones, G.G.; **Kastner, D.W.**; Kulik, H.J. Active Learning Exploration of Transition-Metal Complexes to Discover Method-Insensitive and Synthetically Accessible Chromophores. *JACS Au* **2023**, 3 (2), 391-401. [DOI](#).
3. [\[Front Cover\]](#) Del Rio Flores, A.; **Kastner, D.W.**; Du, T.; Narayanamoorthy, M.; Shen, Y.; Cai, W.; Vennelakanti, V.; Zill, N.A.; Dell, L.B.; Zhai, R.; Kulik, H.J.; Zhang, W. Probing the Mechanism of Isonitrile Formation by a Non-Heme Iron(II)-Dependent Oxidase/Decarboxylase. *JACS* **2022**, 144 (13), 5893-5901. [DOI](#).
4. [\[Front Cover\]](#) Nazemi, A.; Steeves, A.H.; **Kastner, D.W.**; Kulik, H.J. Influence of the Greater Protein Environment on the Electrostatic Potential in Metalloenzyme Active Sites: The Case of Formate Dehydrogenase. *J. Phys. Chem. B* **2022**, 126 (22), 4069-4079. [DOI](#).
5. [\[Front Cover\]](#) Lo, C.L.L.; Joaquin, D.; Moyá, D.A.; Ramos, A.; **Kastner, D.W.**; White, S.A.; Christensen, B.L.; Naglich, J.G.; Degnen, W.J.; Castle, S.L. Synthesis and Evaluation of Potent Yaku'amide A Analogs. *Chem. Sci.* **2022**, 13, 1899-1905. [DOI](#).

Pre-Print and In-Review

6. Colin, Y.K.; **Kastner, D.W.**; Mitchell, J.A.; Gutierrez, M.A.; Yao, S.J.; Neumann, N.E.; Kulik, H.J.; Weng, J. Tracing the Stepwise Darwinian Evolution of a Plant Halogenase. *Science Advances*. **Accepted**.
7. **Kastner, D.W.**; Reinhardt, C.R.; Adamji, H.; Oscar-Okpala, T.S.; Kevlishvili, I.; Román-Leshkov, Y.; Kulik, H. The Role of the Unusual 2-Tyr-1-carboxylate Non-heme Iron Motif in the Mechanism of N,N-Dimethylformamidase. *ChemRxiv*. 2025. [DOI](#).
8. Reinhardt, R.C.; **Kastner, D.W.**; Kulik, H.J. Role of Active Site Residues and Weak Noncovalent Interactions in Substrate Positioning in N,N-Dimethylformamidase. *ChemRxiv*. **2025**. [DOI](#).
9. Keys, A.M.; **Kastner, D.W.**; Kastner, D.W.; Kiessling, L.L.; Kulik, H.J. CH- π Interactions Confer Orientational Flexibility in Protein-Carbohydrate Binding Sites. *ChemRxiv*. **2025**. [DOI](#).
10. Xu, M.; Kastner, D.W.; Luo, W.; Li, F.; Müller, Sun, Y.; Huang, W.; Glinkerman, C.; Guempel, M.; Kulik, H.; Weng, J. Mammalian-like steroidogenesis pathways give rise to cardenolide toxins in seed plants. *Cell* **2025**, In Reviews.

Other Peer-reviewed Publications

11. Keys, A.M.; **Kastner, D.W.**; Kiessling, L.L.; Kulik, H.J. The Energetic Landscape of CH- π Interactions in Protein-Carbohydrate Binding. *Chemical Science*. 2025. [DOI](#).
12. Torrens-Spence, P.M.; Matos, O.J.; **Kastner, D.W.**; Li, T.; Glinkerman, M.C.; Sher, J.; Wang, Y.; Kulik, H.J. Weng, J. Mechanistic Basis for the Emergence of EPS1 as a Catalyst in Plant Salicylic Acid Biosynthesis. *Nat. Commun.* **2024**. [DOI](#).
13. Del Rio Flores, A.; Zhai, R.; **Kastner, D.W.**; Seshadri, K.; Yang, S.; Matias, K.D.; Shen, Y.; Cai, W.; Narayanamoorthy, M.; Do, N.B.; Xue, Z.; Marzooqi, D.A.; Kulik, H.J.; Zhang, W. Enzymatic Synthesis of Azide by a Promiscuous N-nitrosylase. *Nature Chemistry*. **2024**. [DOI](#).
14. Kevlishvili, I.; Vakil, J.; **Kastner, D.W.**; Huang, X.; Craig, S.; Kulik, H. High-Throughput Discovery of Ferrocene Mechanophores with Enhanced Reactivity and Network Toughening. *ChemRxiv*. 2024. [DOI](#).
15. **Kastner, D.W.**; Reinhardt, C.R.; Adamji, H.; Manetsch, M.; Roman-Leshkov, Y.; Kulik, H.J. Dynamic Charge Distribution as a Key Driver of Catalytic Reactivity in an Artificial Metalloenzyme. *ChemRxiv*. **2024**. [DOI](#).

16. Edholm, F.; Nandy, A.; Reinhardt, C.R.; **Kastner, D.W.**; Kulik, H.J. Protein3D: Enabling analysis and extraction of metal-containing sites from the Protein Data Bank with molSimplify. *J. Comput. Chem.* **2023**, *45* (6), 352-361. [DOI](#).
17. Kim, C.Y.; Mitchell, A.J.; **Kastner, D.W.**; Albright, C.E.; Michael, A.G.; Glinkerman, C.M.; Kulik, H.J.; Weng, J. Emergence of a protein exchange-based isomerization and lactonization mechanism in the plant coumarin synthase COSY. *Nat. Commun.* **2023**, *14* (597). [DOI](#).
18. Nandy, A.; Adamji, H.; **Kastner, D.W.**; Vennelakanti, V.; Nazemi, A.; Liu, M.; Kulik, H.J. Using Computational Chemistry to Reveal Nature's Blueprints for Single-Site Catalysis of C–H Activation. *ACS Catal.* **2022**, *12* (15), 9281-9306. [DOI](#).
19. Nandy, A.; Terrones, G.; Arunachalam, N.; Duan, C.; Duan, C.; **Kastner, D.W.**; Kulik, H.J. MOFSimplify, Machine Learning Models with Extracted Stability Data of Three Thousand Metal-Organic Frameworks. *Nat. Sci. Data* **2022**, *9* (74). [DOI](#).
20. Olsen, R.R.; Ireland, A.S.; **Kastner, D.W.**; Groves, S.M.; Spainhower, K.B.; Pozo, K.; Kelenis, D.P.; Whitney, C.P.; Guthrie, M.R.; Wait, S.J.; Soltero, D.; Witt, B.L.; Quaranta, V.; Johnson, J.E.; Oliver, T.G. ASCL1 Represses a SOX9⁺ Neural Crest Stem-like State in Small Cell Lung Cancer. *Genes Dev.* **2021**, *37*, 13-14. [DOI](#).
21. Ireland, A.S.; Micinski, A.M.; **Kastner, D.W.**; Guo, B.; Wait, S.J.; Spainhower, K.B.; Conley, C.C.; Chen, O.S.; Guthrie, M.R.; Soltero, D.; Qiao, Y.; Huang, X.; Tarapcsak, S.; Devarakonda, S.; Chalishazar, M.D.; Gertz, J.; Moser, J.C.; Marth, G.; Puri, S.; Witt, B.L.; Spike, B.T.; Oliver, T.G. MYC Drives Temporal Evolution of Small Cell Lung Cancer Subtypes by Reprogramming Neuroendocrine Fate. *Cancer Cell* **2020**, *38* (1), 60-78. [DOI](#).
22. Joaquin, D.; Lee, M. A.; **Kastner, D. W.**; Singh, J.; Morrill, S. T.; Damstedt, G.; Castle, S. L. Impact of Dehydroamino Acids on the Structure and Stability of Incipient ₃₁₀-Helical Peptides. *J. Org. Chem.* **2020**, *3* (85), 1601-1613. [DOI](#).
23. Morris, D.L.; **Kastner, D.W.**; Johnson, S.; Strub, M.; He, Y.; Bleck, C.K.; Lee, D.; Tjandra, N.; Humanin induces conformational changes in the apoptosis regulator BAX and sequesters it into fibers, preventing mitochondrial outer-membrane permeabilization. *J. Biol. Chem.* **2019**, *50* (294), 19055-19065. [DOI](#).
24. Ashraf, N.M., Krishnagopal, A., Hussain, A., **Kastner, D.W.**, Sayed, A.M., Mol Y.K., Swaminathan, K., Zeeshan, N. Engineering of serine protease for improved thermostability and catalytic activity using rational design. *Int. J. Biol. Macromol.* **2018**, *126*, 229-236. [DOI](#).
25. Jalan, A.; **Kastner, D.W.**; Webber, K.G.I.; Smith, M.S.; Price, J.L.; Castle, S.L. Bulky dehydroamino acids enhance proteolytic stability and folding in β -hairpin peptides. *Org. Lett.* **2017**, *19* (19), 5190-5193. [DOI](#).
26. Ashraf, N.M.; Imran, K.; **Kastner, D.W.**; Ikram, K.; Mushtaq, A.; Hussain, A.; Zeeshan, N. Potential involvement of mi-RNA 574-3p in progression of prostate cancer: A bioinformatic study. *Mol. Cell. Probes* **2017**, *36*, 21-28. [DOI](#).

Other Publications

27. **Kastner, D.W.** Human Processing Unit. *Sci. Phi. J.* **2024**, *2*, 53-55. [ISSN 2205-0310](#).
28. **Kastner, D.W.** Computational Modelling of Peptides Containing Non-Standard Amino Acids *Undergraduate Honors Thesis* **2019**. *ScholarsArchive*, 61. [ISSN](#).
29. **Kastner, D.W.** The Ultimate Triumph of Truth. *Brigham Young University Commencement*, **2019**. URL: speeches.byu.edu.

Selected Presentations

1. **Kastner, D.W.**; Clorice, R.; Adamji, H.; Manetsch, M.; Kulik, H.J. Data-driven insights from dynamic electronic properties of artificial metalloenzymes. Oral Presentation. *ACS Fall Conference Denver*, **2024**.

2. **Kastner, D.W.** and Kulik, H.J. Blueprints of C–H bond activation across iron-containing high-valent metalloenzymes. Poster Presentation. *ACS Fall Conference Denver*, **2024**.
3. **Kastner, D.W.**; Clorice, R.; Adamji, H.; Manetsch, M.; Kulik, H.J. Mechanistic Insights from Electronic Properties in Artificial Mini-enzymes. Poster Presentation. *ACS Spring Conference Denver*, **2024**. [DOI](#).
4. **Kastner, D.W.**; Nandy, A.; Mehmood, R.; Kulik, H.J. Substrate positioning in Non-heme Fe(II)/ α -ketoglutarate-dependent halogenases and hydroxylases: A computational study. Oral Presentation. *Virtual Conference on Chemistry and its Applications (VCCA)*, **2023**. [Video recording](#).
5. **Kastner, D.W.**; Jalan, A.; Castle, S. L. Conformational ensemble calculations of proteolytically stable β -hairpins containing bulky α,β -dehydroamino acids. *American Chemical Society 254th National Meeting*, Washington D.C., **2017**.
6. **Kastner, D.W.**; and Castle, Steven L., Progress toward synthetically simplified natural anticancer peptide (2018). *Library Undergraduate Poster Competition 2018*, 5. [ISSN](#)
7. **Kastner, D.W.**; Castle, S. L. Computational predictions β -hairpins containing bulky dehydroamino acids. *Scholars Archive 2017*, 4. [ISSN](#)
8. **Kastner, D.W.**; Lo, C. C. L.; Castle, S. L. Progress towards a synthetically simplified anticancer peptide. *Student Research Conference (SRC)*, Provo, **2018**.
9. **Kastner, D.W.**; Jalan, A.; Castle, S. L. QM/MM analysis of proteolytically stable β -hairpins. *Student Research Conference (SRC)*, Provo, **2017**.
10. **Kastner, D.W.**; Castle, S. L. ONIOM geometry optimization of bulky dehydroamino acids in β -hairpins. *IEEE Poster Session*, Provo, **2017**.

Advanced Coursework and Instruction

Massachusetts Institute of Technology

<i>Biological Engineering</i> • Fundamentals of Drug Development (20.201)	Grade: A (5.0)
<i>Biological Engineering</i> • Research Experience in Biopharma (20.930)	Grade: A (5.0)
<i>Biological Engineering</i> • Protein Engineering (20.535)	Grade: A (5.0)
<i>Biological Engineering</i> • Analysis of Biological Networks (20.440)	Grade: A (5.0)
<i>Computer Science</i> • Modeling with Machine Learning: Algorithm to Apps (6.C51)	Grade: A (5.0)
<i>Chemical Engineering</i> • Machine Learning for Molecular Engineering (10.C51)	Grade: A (5.0)
<i>Chemical Engineering</i> • Computational Chemistry (10.637)	Grade: A (5.0)

Brigham Young University

<i>Physiology</i> • Biophysics (PDBIO 568)	Grade: A (4.0)
<i>Physiology</i> • Advanced Physiology (PDBIO 362)	Grade: A (4.0)
<i>Biology</i> • Cell Biology (BIO 360)	Grade: A (4.0)
<i>Biology</i> • Computational Biology (BIO 362)	Grade: A (4.0)
<i>Chemistry</i> • Biophysical Chemistry (CHEM 468)	Grade: A (4.0)
<i>Chemistry</i> • Organic Chemistry (CHEM 351 & 352)	Grade: A (4.0)
<i>Chemistry</i> • Biochemistry (CHEM 481)	Grade: A (4.0)
<i>Physics</i> • Molecular Dynamics (PDBIO 550R)	Grade: A (4.0)
<i>Physics</i> • Electricity and Magnetism (PHSCS 220)	Grade: A (4.0)
<i>Physics</i> • Newtonian and Modern Physics (PHSCS 121 & 123)	Grade: A (4.0)

Teaching Experience

MIT Instructor • <i>Biological Engineering</i> • 3D Scientific Rendering (20.S947)
MIT Teaching Assistant • <i>Chemical Engineering</i> • Computational Chemistry (10.437)
MIT Teaching Assistant • <i>Biological Engineering</i> • Principles of Molecular Bioengineering (20.420)
BYU Teaching Assistant • <i>Cell Biology & Physiology</i> • Cell Biology (PDBIO 360)

Current Societal and Honors Affiliations

Foundation for Science and Disability (FSD)
Society for Advancement of Chicanos/Hispanics & Native Americans in Science (SACNAS)
National Spanish Speaker Honor Society (SDP)
American Chemical Society (ACS) – Computers in Chemistry (COMP) Division
The Protein Society (TPS)
National Scientific Research Honor Society Sigma Xi

Press

[“The art of the enzyme”](#) MIT News, **2024**.
[“Computational model captures the elusive transition states of chemical reactions”](#) MIT News, **2023**.
[“Scientists use computational modeling to design “ultrastable” materials”](#) MIT News, **2023**.
[“Computational modeling guides development of new materials”](#) MIT News, **2022**.
[“The Ultimate Triumph of Truth”](#) BYU Speeches, **2019**.
[“Part of the search: Q&A with BYU Commencement Speaker”](#) BYU Honors Program News, **2019**.
[“A wonderful world to share joy, show love, create connections”](#) BYU News, **2019**.
[“Kastner chosen as commencement speaker at BYU”](#) Napa Valley Register, **2019**.
[“Rex Lee Run unites those affected and impacted by cancer”](#) Daily Herald, **2018**.
[“Kastner receives fellowship for cancer research”](#) Napa Valley Register, **2018**.
[“Napa’s Kastner family creates artwork for new dealership”](#) Napa Valley Register, **2016**.